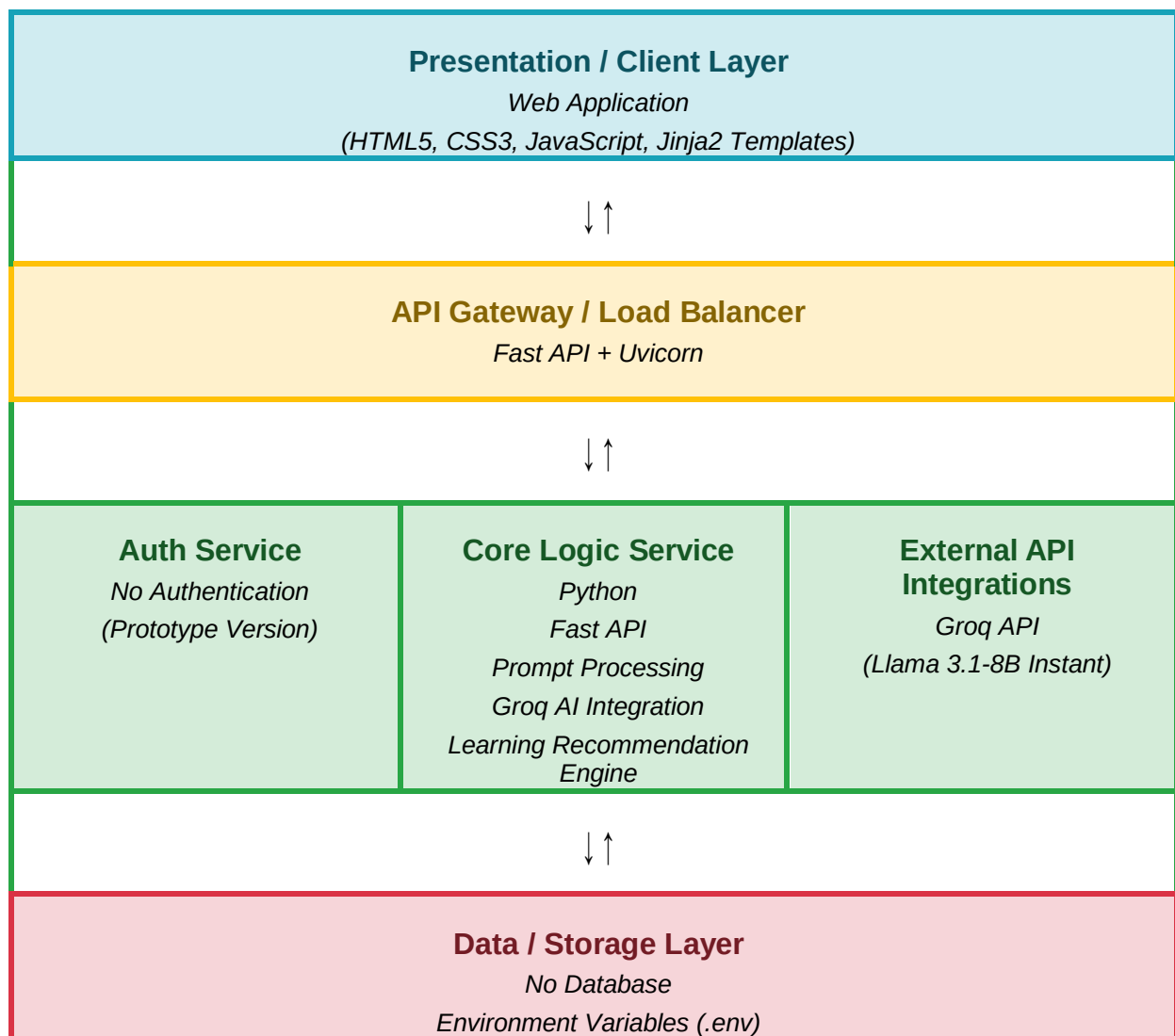


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface where students can enter questions, choose learning tasks, and view AI-generated responses.	HTML5, CSS3, JavaScript, Jinja2
API Gateway	Receives user requests, routes them to the backend, and returns responses to the client.	Fast API, Uvicorn
Core Logic Service	Validates user input, builds prompts, communicates with the Groq AI model, and formats educational responses.	Python, Fast API, OpenAI SDK
External API Integration	Connects the application to the Groq AI service for generating educational content.	Groq API (Llama 3.1-8B Instant)
Data / Storage Layer	Stores application configuration such as API keys. The prototype does not use a database for user data.	Python-dotenv (.env)